

**WEAVER: A SPATIO-TEMPORAL DEEP LEARNING MODEL ARCHITECTURE
BASED ON THE MIXED KRONECKER MATRIX-VECTOR IDENTITY**

Christopher Cheong, M.S.

Department of Civil, Environmental, and Geo- Engineering
University of Minnesota, Twin Cities
cheon028@umn.edu

Seongjin Choi, Ph.D., Corresponding Author

Department of Civil, Environmental, and Geo- Engineering
University of Minnesota, Twin Cities
chois@umn.edu

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1 ABSTRACT

2 Intelligent Transportation Systems demand sophisticated traffic forecasting models that can process
3 massive data streams and generate accurate predictions, allowing optimizations in traffic flow and
4 anomaly detection. This work proposes the Weaver model, a novel spatio-temporal deep learning
5 architecture that is relatively uncomplicated, quick to train and deploy, while demonstrating excellent
6 traffic forecast capabilities. Unlike Transformers, the Weaver does not require specially formulated
7 positional encodings to encode sequence order, nor does it rely on stacked self-attention mechanisms.
8 Instead, the Weaver keeps spatial and temporal dimensions separate, allowing activation maps to
9 correspond directly to each dimension. These activation maps are generated using the Tanimoto
10 coefficient, which enables rich encodings of intra-dimensional associations. This approach creates
11 nuanced spatio-temporal representations bounded on the interval $[-0.333, 1]$, showing potential
12 for interpretability while contributing to the model's unique architecture. The Weaver model thus
13 strikes an effective balance between model complexity, computational efficiency, and forecasting
14 accuracy, offering a promising solution for advanced traffic prediction in Intelligent Transportation
15 Systems.

16
17 *Keywords:* Weaver Model, Spatio-Temporal Forecasting, Traffic Forecasting, Network Forecast-
18 ing, Deep Learning, mixed Kronecker Matrix-Vector Identity, Roth's Column Lemma, Tanimoto
19 coefficient, Interpretable Activation Map, Transformer

1 INTRODUCTION

2 While transportation has evolved from the Oregon Trail days to complex urban networks monitored
 3 by extensive sensor systems, the focus of modern traffic management lies in leveraging these vast
 4 spatio-temporal datasets with advanced deep learning techniques. Initially, graph-based methods
 5 like Graph Wavenet (1), DCRNN (2), and STGCN (3) were favored due to their natural fit with
 6 traffic network structures. However, recent advancements have seen the adaptation of Transformer
 7 models (4), originally developed for NLP, to numerical spatio-temporal data prediction. This
 8 shift has given rise to specialized Transformer variants for traffic forecasting, including STTN (5),
 9 STAEformer (6), Traffic Transformer (7), and Spacetimeformer (8), and ASTGCRN (9).

10 Deep learning models outperform classical methods like Kalman filters and ARIMA in
 11 accuracy, but often face scalability issues. While techniques such as graph pruning and attention
 12 module approximation partially address these challenges, the evolving landscape of traffic networks
 13 demands more. As modern networks transition into intelligent transportation systems with rapid
 14 data collection, there's a growing need for models that train quickly, require less storage, and
 15 deploy faster. Despite advancements in data processing, relying solely on hardware improvements
 16 to manage model complexity is unsustainable. This shift creates opportunities for innovative
 17 approaches that balance accuracy with efficiency in the new era of smart transportation.

18 Traffic networks can be viewed as fully connected systems where events in one location
 19 and time can influence distant parts of the network later. A study of Paris's traffic from 2014-
 20 2018 supports this, suggesting that congestion waves propagate like a diffusion process with a
 21 distance-decaying correlation function. During peak hours, this correlation length diverges, allowing
 22 congestion effects to spread across the entire network (10). Similar findings have been reported in
 23 other regions (11, 12).

24 While modeling latent spatio-temporal associations within an all-to-all time-expanded net-
 25 work framework is plausible (13), directly applying vanilla Transformer architecture results in
 26 prohibitive $\mathcal{O}(t^2n^2)$ complexity. Strategies to mitigate this include separate processing of spatial
 27 and temporal data views, kernel approximations for attention modules, and selective attention using
 28 graph information. However, despite these adaptations, fundamental interpretability challenges per-
 29 sist in spatio-temporal Transformers. Although they split data into spatial and temporal components
 30 (5, 6), the Transformer's reliance on stacked attention modules creates a complex web of inter-
 31 dependencies within the latent space. This succession of attention mechanisms makes it difficult to
 32 pinpoint the precise chain of operations contributing to a particular output, thus complicating the
 33 interpretation of model behavior and decision-making processes in the context of spatio-temporal
 34 forecasting.

35 Furthermore, the necessity of positional embeddings in Transformer-based models that are
 36 naturally order-invariant alludes to the dependence of model performance on the form of positional
 37 embeddings used. For most applications, the vanilla positional embeddings introduced in (4) may
 38 suffice, there is an increasing body of research that suggest alternative formulations can affect
 39 Transformer performance (14–16). This trend can become problematic, as it fosters an intensive
 40 diversification of positional embedding formulations, each micro-engineered to excel in specific
 41 domains. Ideally, model performance should not depend on the type of sequence markers used, but
 42 rather on representations derived from real-world information and data quality.

43 To address these challenges, we propose the *Weaver*, a novel spatio-temporal deep learning
 44 architecture designed to capture cross-dimensional associations while maintaining distinction
 45 between spatial and temporal dimensions. The *Weaver* leverages the Kronecker Matrix-Vector

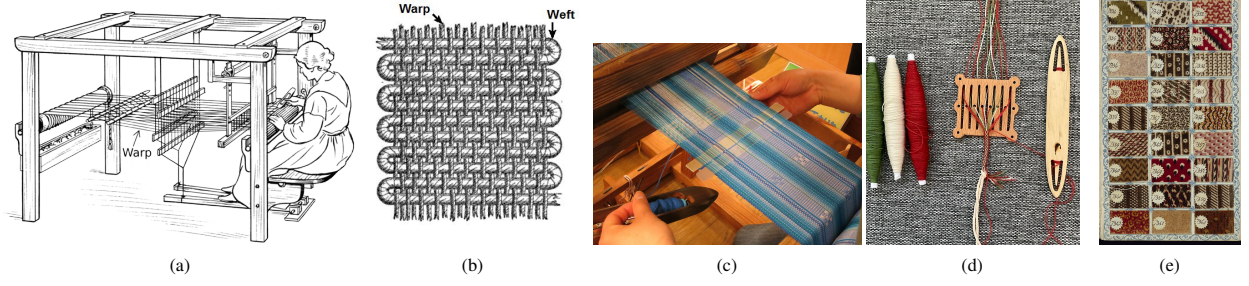


FIGURE 1: (a) Weaver operating treadle loom, holding shuttle in right hand, with warp yarns taut between warp and cloth beams. (b) Illustration of static warp yarns and interwoven weft yarns in fabric. (c) Traditional minsa textile weaving on Ishigaki Island, showing shuttle passing weft through taut warp yarns. (d) Rigid heddle controlling colored warp yarns for pattern weaving. (e) Page from 1784 English salesman's sample book showing fabric swatches. See Acknowledgements for image attributions and licensing.

product and the Tanimoto coefficient to effectively model spatio-temporal relationships. Our key contributions include a new approach to spatio-temporal modeling that maintains dimensional interpretability, a scalable architecture that performs competitively in traffic forecasting tasks while requiring less computational resources, and a model design that enhances traceability between inputs and outputs. Unlike Transformers, which use scaled dot-product attention, softmax, and positional embeddings, the *Weaver* employs the Tanimoto coefficient for more interpretable spatio-temporal relationship modeling without relying on positional information.

The *Weaver*'s namesake and its components draw inspiration from a weaver's meticulous work on a loom (Fig. 1a). This process involves translating designs from a weaving draft onto fabric by interlacing lateral weft yarns through static longitudinal warp yarns (Fig. 1b) using a shuttle. The heddle (Fig. 1d) separates warp yarns to facilitate pattern creation, while swatches (Fig. 1e) serve as small fabric samples showcasing the finished textiles.

LITERATURE REVIEW

PRELIMINARIES

In this section, we introduce key concepts, tensor operations, and the problem definition, then detail the graph structure used in designing the *Weaver* deep learning spatio-temporal forecasting model. We follow with a description of the *Weaver*'s modules and their operating principles. For brevity, we exclude the batch size dimension from our notation.

Key Concepts

Key concepts specific to this work are defined here. For a general overview of standard deep learning terminologies and structures, we refer to (17).

1. **Kronecker product:** The Kronecker product of two matrices $\mathbf{R} \in \mathbb{R}^{P \times Q}$ and $\mathbf{S} \in \mathbb{R}^{M \times N}$ is denoted by $\mathbf{R} \otimes \mathbf{S}$. The result is a block matrix of size $(P \cdot M) \times (Q \cdot N)$, where each element r_{ij} of matrix $\mathbf{R}$ is multiplied by the entire matrix $\mathbf{S}$. Formally, the Kronecker product is defined as:

$$\mathbf{R} \otimes \mathbf{S} = \begin{bmatrix} r_{00}\mathbf{S} & \cdots & r_{0(Q-1)}\mathbf{S} \\ \vdots & \ddots & \vdots \\ r_{(P-1)0}\mathbf{S} & \cdots & r_{(P-1)(Q-1)}\mathbf{S} \end{bmatrix} \quad (1)$$

where each block $r_{ij}\mathbf{S}$ represents the element r_{ij} of matrix $\mathbf{R}$ multiplied by the entire matrix $\mathbf{S}$.

2. Kronecker Matrix-Vector Identity (KMV): The Kronecker Matrix-Vector Identity (18) states that for matrices $\mathbf{R} \in \mathbb{R}^{P \times Q}$, $\mathbf{S} \in \mathbb{R}^{Q \times R}$, $\mathbf{K} \in \mathbb{R}^{R \times S}$,

$$(\mathbf{K}^\top \otimes \mathbf{R}) \text{vec}(\mathbf{S}) = \text{vec}(\mathbf{RSK}), \quad (2)$$

where vec is the vectorization operator.

For more information on the properties of the KMV and Kronecker products, please refer to (18) and (19).

3. Tanimoto Similarity Coefficient, Θ :

$$\Theta(\mathbf{r}, \mathbf{s}) = \frac{\mathbf{r} \cdot \mathbf{s}}{\|\mathbf{r}\|^2 + \|\mathbf{s}\|^2 - \mathbf{r} \cdot \mathbf{s}}. \quad (3)$$

Tensor Notation and Operations

For general tensor notation, this work follows (20).

During tensor operations, the subscript brackets $\mathcal{A}_{[]}$ show the current dimensional arrangements of the tensor, e.g. $\mathcal{A}_{[I_0, I_1, \dots, I_{\eta-1}]} \equiv (\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{\eta-1}})$. Additionally, $*$ will be used as shorthand for all arbitrary dimensions preceding the stated end-dimensions, e.g., $(\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{\eta-3} \times J \times K}) \equiv (\mathcal{A} \in \mathbb{R}^{* \times J \times K})$.

This work uses the following overrides for some common matrix notations and operations:

1. Tensor dimension and iterator indexing:

Given a tensor $\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{\eta-1}}$, the indices of $\mathcal{A}$ are defined as: $\mathcal{A}_{i_0 i_1 \dots i_{\eta-1}}$, where each $i_\xi : \xi \in \{0, 1, \dots, \eta-1\}$ represents the iterator along the ξ -th dimension of the tensor. The range of each index i_ξ is $0 \leq i_\xi < I_\xi$ to be consistent with Python indexing protocols.

2. Batched tensor transpose:

The typical transpose notation $\mathcal{A}^\top$ shall become the shorthand for tranpose that only operate along the last 2 dimensions when applied to tensors, i.e., if an arbitrary tensor $\mathcal{A} \in \mathbb{R}^{* \times J \times K}$, then $\mathcal{A}^\top \in \mathbb{R}^{* \times K \times J}$.

3. Batched tensor multiplication:

"Tensor multiplication" in this work refers to the batched tensor multiplication as defined here. Given two tensors $\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{(\eta-3)} \times Q \times J}$ and $\mathcal{B} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{(\eta-3)} \times K \times J}$, the batched tensor multiplication operates only along the last two dimensions. Moreover, tensors $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ must share the leading dimensions $I_0 \times I_1 \times \dots \times I_{(\eta-3)}$.

$$\mathcal{C} = \mathcal{A} \cdot \mathcal{B} = \mathcal{AB}^\top \text{ where } \mathcal{C} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{(\eta-3)} \times Q \times K}. \quad (4)$$

The tensor multiplication is defined as the operation:

$$\mathcal{C}_{i_0 i_1 \dots i_{(\eta-3)} q k} = \sum_j \mathcal{A}_{i_0 i_1 \dots i_{(\eta-3)} q j} \cdot \mathcal{B}_{i_0 i_1 \dots i_{(\eta-3)} j k}. \quad (5)$$

4. Batched Kronecker product:

The Kronecker product in this work operates along the last 2 dimensions of a tensor. Given two tensors that share all dimensions prior to the last two: $\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{(\eta-3)} \times Q \times J}$ and $\mathcal{B} \in \mathbb{R}^{I_0 \times I_1 \times \dots \times I_{(\eta-3)} \times K \times L}$, the Kronecker product of $\mathcal{A}$ and $\mathcal{B}$, denoted by $\mathcal{C} = \mathcal{A} \otimes \mathcal{B}$, is a tensor $\mathcal{C} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_{(\eta-3)} \times (Q \cdot K) \times (J \cdot L)}$ defined as:

$$\mathcal{C}_{i_1 i_2 \dots i_N (qK+k)(jL+\ell)} = \mathcal{A}_{i_1 i_2 \dots i_N i j} \cdot \mathcal{B}_{i_1 i_2 \dots i_N k \ell}, \quad \text{for all valid indices } i_1, i_2, \dots, i_N, q, j, k, \ell. \quad (6)$$

1 Some elementary dimensional manipulation operations are as follows:

2 1. **Rearrangement:**

3 Tensor dimensional rearrangement, necessary for aligning dimensions in downstream operations,
4 is denoted by $\longrightarrow$ with the specific rearrangement type indicated above the arrow.

5 2. **Transpose:**

6 Dimensional transpose beyond the last two dimensions are denoted by $\leftrightarrow$ during dimensional
7 manipulation.

8 3. **Merge:**

9 Dimensional merge (or "flattening") is denoted by $A \cdot B$.

10 4. **Split:**

11 Splitting of a tensor dimension is denoted by $A \rightarrow B \times C$, wherein $A = B \times C$ and $A, B, C \in$
12 $\{\mathbb{N}_+ \setminus \infty\}$.

13 The compound tensor manipulations are as follows:

14 1. **MultiHead(.):**

15 The multi-head concept is adapted from Transformers to allow the *Weaver* to learn a variety of
16 association patterns from the data. Let $\mathcal{A}$ be an arbitrary tensor, $\mathcal{A} \in \mathbb{R}^{*\times C}$, where $\eta \geq 0$. The
17 $\text{MultiHead}(H)(\mathcal{A})$ operation applies only along the end dimension of $\mathcal{A}$, C at $\xi = (\eta - 1)$.

$$18 \text{MultiHead}(H)(\mathcal{A}) := \left[\mathcal{A}_{[*],C} \xrightarrow{C \rightarrow H, \frac{C}{H}} \mathcal{A}_{[*],H, \frac{C}{H}} \xrightarrow{*\leftrightarrow H} \mathcal{A}_{[H,*], \frac{C}{H}} \right], \quad (7)$$

20 2. **Linear(.):**

21 Linear layers are replaced by the $\text{Linear}(I, O)(.)$ operation with weight tensors $\mathbf{W}$, and bias
22 tensor $\mathbf{B}$. The $\text{Linear}(I, O)(.)$ operation only operates on the last dimension of the tensor. For
23 $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_{(\eta-2)} \times R}$, where $\eta \geq 0$:

$$24 \text{Linear}(R, S)(\mathcal{A}) := (\mathcal{A}\mathbf{W} + \mathbf{B}) \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_{(\eta-2)} \times S}, \quad (8)$$

25 where $\mathbf{W} \in \mathbb{R}^{R \times S}$, $\mathbf{B} \in \mathbb{R}^{* \times S}$.

26 3. **Expand(.):** (TODO: To tensor and check dimensions)

27 The $\text{Expand}(\xi, \kappa)(\mathcal{A})$ operation adds an extra dimension of size 1 along index ξ , then replicates
28 the tensor along ξ by κ times. For example, applying $\text{Expand}(\cdot)$ to $\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times I_2}$ along $\xi = 1$
29 by κ times yields the following:

$$30 \text{Expand}(\xi = 1, \kappa)(\mathcal{A}) = [\mathcal{A}_{[I_0, I_1, I_2]} \rightarrow \mathcal{A}_{[I_0, 1, I_1, I_2]} \rightarrow \mathcal{A}_{[I_0, \kappa, I_1, I_2]}]. \quad (9)$$

31 **Problem Definition**

32 A traffic network can be represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$. The vertex set $\mathcal{V}$ contains
33 nodes corresponding to sensors at key network points like intersections or ramps, measuring data
34 such as traffic volume or speed. The edge set $\mathcal{E}$ comprises links between nodes, either connected by
35 roadways or associated by proximity. The adjacency matrix $\mathbf{A}$ contains link weights A_{ij} , which can
36 represent various metrics like Manhattan distance, Euclidean distance, or travel time between nodes.

37 In traffic forecasting, deep learning models are used to learn and generate predictions of
38 traffic measures $\mathbf{X}_t$, typically the traffic speed, flow, or volume. Given a sequence from a time
39 horizon P of past observations $\{\mathbf{X}_{t-P+1}, \dots, \mathbf{X}_t\}$, a corresponding sequence of predictions is made
40 for a horizon of Q timesteps in the future: $\{\mathbf{X}_{t+1}, \dots, \mathbf{X}_{t+Q}\}$.

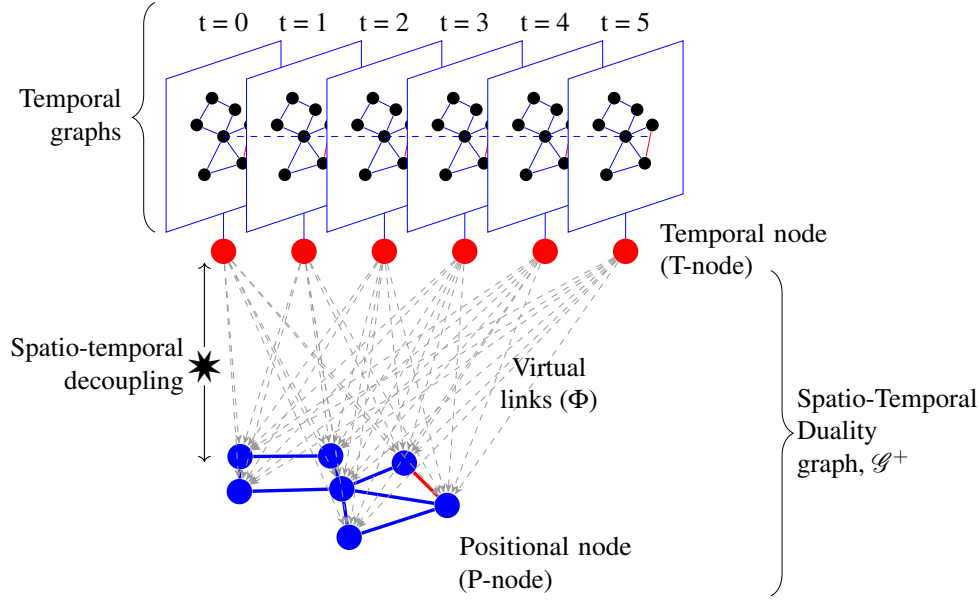


FIGURE 2: Illustration of the Spatio-Temporal Duality graph, the decoupling of classical temporal graphs into separate temporal (T-node) and spatial (P-node) components to isolate time-sensitive and location-sensitive information.

1 The Spatio-Temporal Duality Graph, $\mathcal{G}^+$

2 We extend the definition of $\mathcal{G}$ by introducing time steps as distinct node-entities, separating temporal
3 and locational aspects in traffic data. We replace the classical definition of nodes with the following
4 two node sub-types, as shown in Fig. 2:

- 5 1. *Positional-nodes (P-nodes)*, ρ , associated with sensor locations in space and carrying location-
6 dependent information. For instance: signalized intersections, central business districts (CBD),
7 and speed limit zones.
- 8 2. *Temporal-nodes (T-nodes)*, τ , associated with specific points in time and carrying time-dependent
9 information. For instance: time of day, day of week, and holidays.

10 T-nodes connect to P-nodes via directed "virtual links" (Fig. 3a), replacing weighted links.
11 These virtual links represent projection functions that overlay time-dependence on the physical
12 system, modifying P-node states based on T-node temporal information. The resulting Spatio-
13 Temporal Duality graph in Fig. 4d is $\mathcal{G}^+ = \{\mathcal{V}^+, \mathcal{E}^+, \mathbf{P}, \mathbf{T}\}$, where

- 14 • $\mathcal{V}^+ = \{(\tau_t, \mathcal{V}_P) : t < t + 1\}$ is the Spatio-Temporal Duality vertex set containing paired tuples
15 of T-nodes representing time point t , and the entire P-node vertex set $\mathcal{V}_P = \{\rho_0, \rho_1, \dots, \rho_N\}$
16 containing all P-nodes in the network. $\mathcal{V}_T = \{\tau_P, \tau_{P+1}, \dots, \tau_t, \tau_{t+1}, \dots, \tau_{Q-1}, \tau_Q : t < t + 1\}$ is
17 the vertex subset containing only all instances of T-nodes τ , such that $\mathcal{V}^+ = (\mathcal{V}_P \cup \mathcal{V}_T)$ and
18 $(\mathcal{V}_P \cap \mathcal{V}_T) = \emptyset$. $|\mathcal{V}_P| = N$, $|\mathcal{V}_T| = T$, and $|\mathcal{V}^+| = |\mathcal{V}_P| + |\mathcal{V}_T|$.
- 19 • $\mathbf{P} \in \mathbb{R}^{N \times N}$ is the positional adjacency matrix, describing a measure of association between two
20 locations, ρ_i and ρ_j .
- 21 • $\mathbf{T} \in \mathbb{R}^{T \times T}$ is the temporal perturbation matrix containing link weights describing the measure of
22 impact from temporal changes in the system when transitioning from τ_i to τ_j . Let $T_{ij} \in (\mathbf{T} \cap \mathbb{R})$
23 be the value at (τ_i, τ_j) .
- 24 • $\mathcal{E}^+$ is the Spatio-Temporal Duality edge set containing three distinct sets of links:

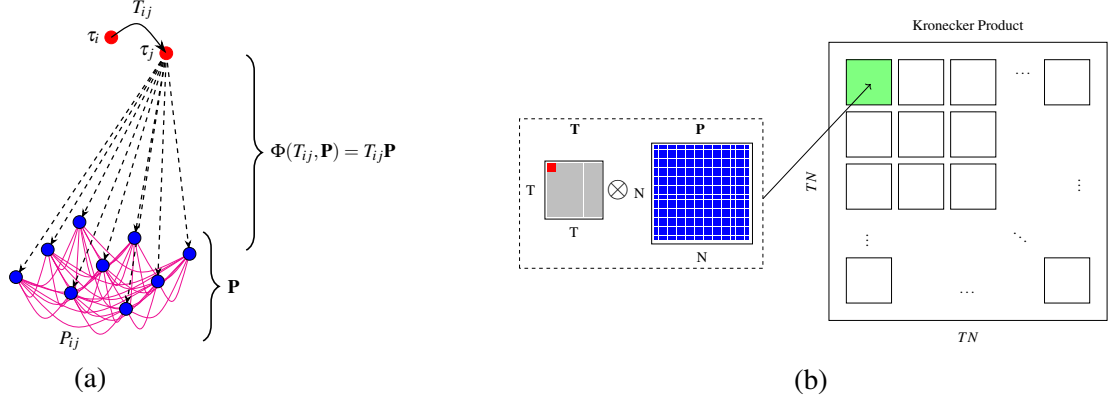


FIGURE 3: (a) Illustration of the Kronecker structure in $\mathcal{G}^+$. $T_{ij}\mathbf{P}$ is a submatrix of $(\mathbf{T} \otimes \mathbf{P})$, where $T_{ij} \in \mathbf{T}$ is the temporal link weight from T-node i to j , and $\mathbf{P}$ is the spatial adjacency matrix. Red nodes represent temporal nodes (T-nodes), and blue nodes represent positional nodes (P-nodes). **(b)** An equivalent Kronecker representation of the graph structure in (a).

- 1 $-\mathcal{E}_P = \{(\rho_i, \rho_j) : \rho_i, \rho_j \in \mathcal{V}_P\},$
- 2 $-\mathcal{E}_T = \{(\tau_i, \tau_j) : \tau_i, \tau_j \in \mathcal{V}_T\},$ temporal information is allowed to travel both ways, so no specific
- 3 ordering is required.
- 4 $-\mathcal{E}_{PT} = \{\varphi(T_{ij}, \mathbf{P}) : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}^{N \times N}\}.$

5 Defining the mapping function as: $f(T_{ij}, \mathbf{P}) = T_{ij} \cdot \mathbf{P}$, we can thus write the mapping function

6 for the entire network as

$$\Phi(\mathbf{T}, \mathbf{P}) = (\mathbf{T} \otimes \mathbf{P}). \quad (10)$$

8 This mapping assumes uniform distribution of temporal information across all P-nodes,

9 implying homogeneous time-sensitive events throughout the network, but this assumption may

10 not be entirely realistic.

11 To address this limitation, a *multi-head activation* design adapted from Transformers can

12 be employed, allowing each activation head to capture different aspects of time-sensitive

13 information before combination, thus providing a more nuanced representation of temporal

14 dynamics across the network.

15 Three distinct subgraphs exist in $\mathcal{G}^+$ containing spatial, temporal or spatio-temporal compo-

16 nents of $\mathcal{G}^+$:

- 17 1. $\mathcal{G}_P^+ = \{\mathcal{V}_P, \mathcal{E}_P, \mathbf{P}\},$
- 18 2. $\mathcal{G}_T^+ = \{\mathcal{V}_T, \mathcal{E}_T, \mathbf{T}\},$
- 19 3. $\mathcal{G}_{TP}^+ = \{\mathcal{V}^+, \mathcal{E}_{TP}, \mathbf{P}, \mathbf{T}\}.$

21 Consequently, we designed a module based on the Tanimoto Coefficient for learning intra-

22 dimensional link weight matrices $\mathbf{P}$ and $\mathbf{T}$ in $\mathcal{G}_P^+$ (Fig. 4a) and $\mathcal{G}_T^+$ (Fig. 4b), and a module based

23 on the Kronecker Matrix-Vector (KMV) product for learning inter-dimensional link interactions

24 (Fig. 4c) assuming the mapping function form in Eqn. (10).

25 THE WEAVER: A SPATIO-TEMPORAL FORECAST MODEL

26 Overview

27 The *Weaver* (Fig. 5) is a novel, simple, and interpretable model architecture for spatio-temporal

28 data analysis. It leverages the Kronecker Matrix-Vector Identity (KMV) to learn and distinguish in-

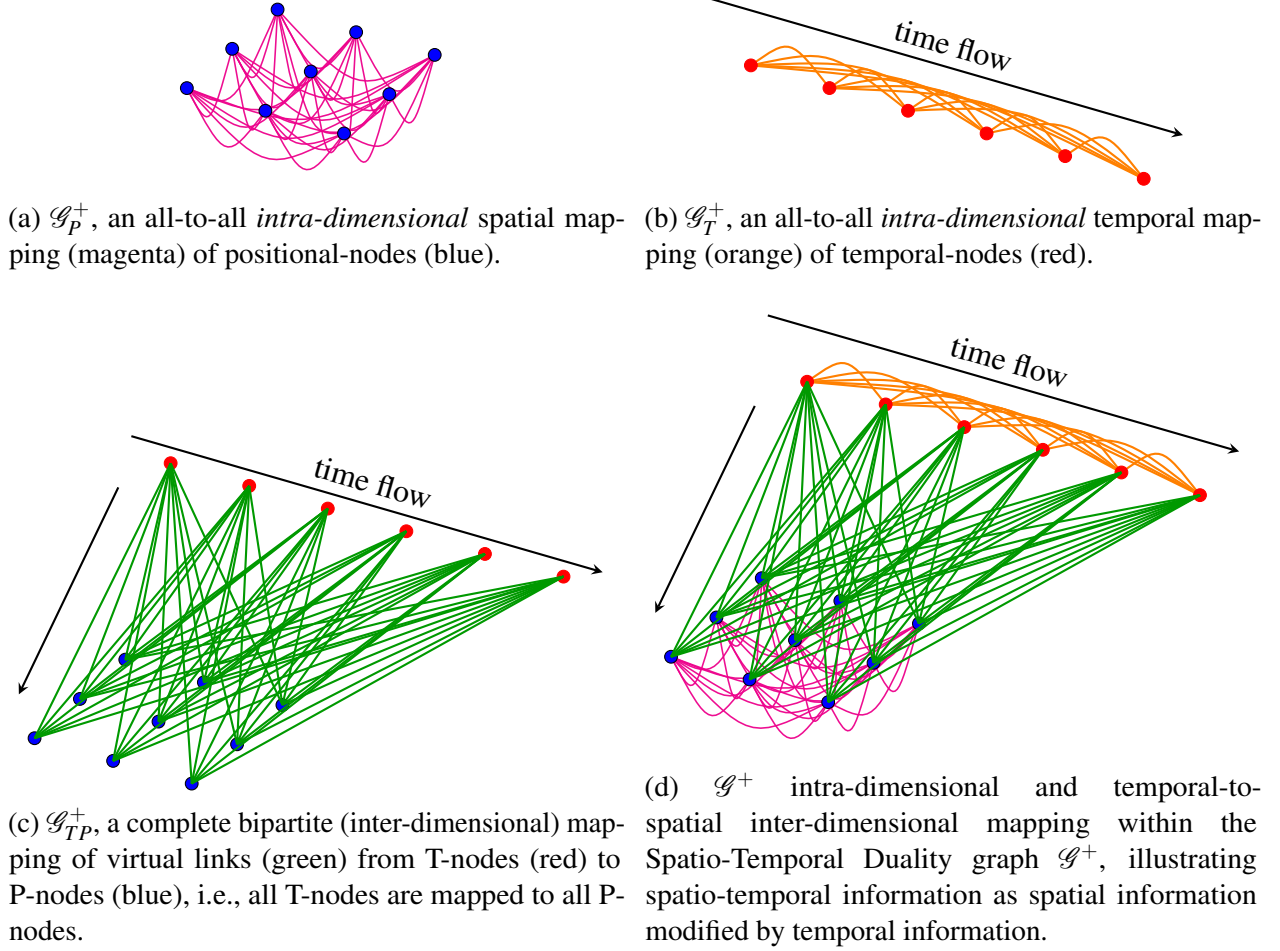


FIGURE 4: Illustration of graphs $\mathcal{G}^+$ and subgraphs: $\mathcal{G}_P^+$, $\mathcal{G}_T^+$, $\mathcal{G}_{TP}^+$ for temporal nodes (T-nodes) and positional nodes (P-nodes) with example connections using 6 T-nodes and 9 P-nodes.

1 terwoven patterns while maintaining separate spatial and temporal dimensions. Using the Tanimoto
 2 coefficient as its core metric for intra-dimensional associations, the *Weaver* achieves state-of-the-art
 3 prediction performance. While capable of learning directly from data, it can also incorporate exoge-
 4 nous spatio-temporal information when available, enhancing model convergence and prediction
 5 accuracy.

6 Unlike Transformer-based models (4) that rely on stacked attention modules, the *Weaver*
 7 centralizes inference in the *Loom* module, with peripheral modules like *Swatch* and *Heddle* encoders
 8 not actively participating in inference. This structural consistency allows precise identification of
 9 dimensional representation formation: the *WarPer* processes the temporal view, the *WefTer* handles
 10 the spatial view, and the *Shuttle* interweaves these outputs for spatio-temporal interactions. This
 11 modular design enhances interpretability by delegating specific information types to dedicated
 12 components.

13 First in the *Weaver* pipeline are two encoders (Fig. 6): the *Swatch* and *Heddle*. The *Swatch*
 14 augments the dataset by extracting intrinsic information within the dataset, whereas the *Heddle*
 15 processes and appends information from exogenous covariates to augment the dataset. Both the
 16 *Swatch* and *Heddle* do not need to be utilized concurrently, but the *Weaver* works best when one or

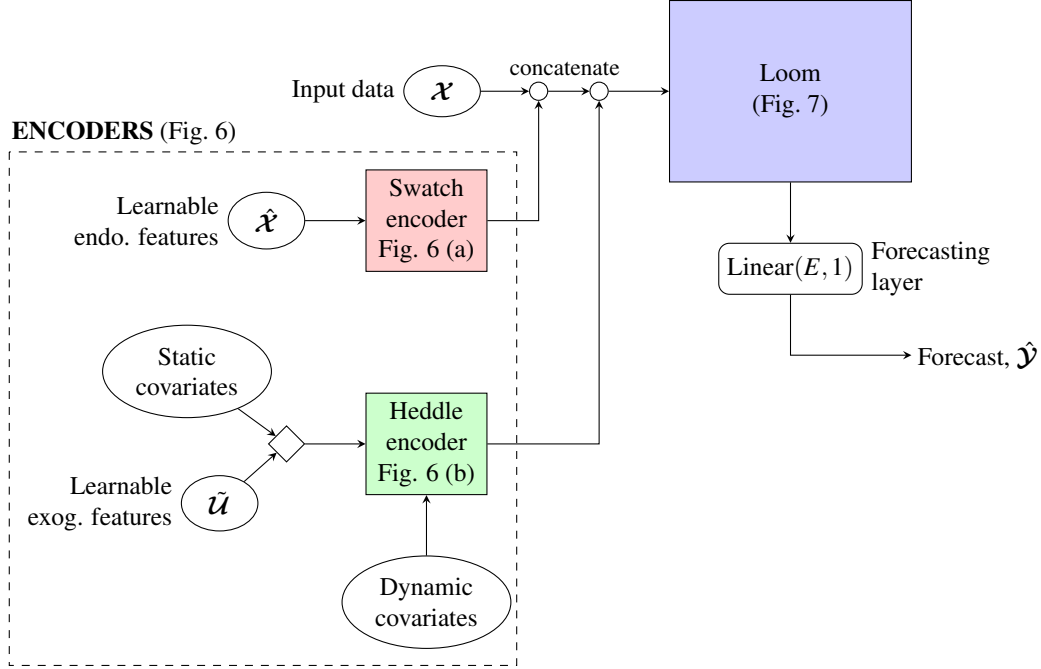


FIGURE 5: An overview of the architecture of the *Weaver*, a spatio-temporal forecast model. The *Swatch* is shown in red, the *Heddle* in yellow and the *Loom* in purple.

the other is active.

1. **Swatch encoder** (Fig. 6a): The *Swatch* allows the *Weaver* to self-augment the input data by learning and utilizing static latent representations inherent to the spatio-temporal data.
2. **Heddle encoder** (Fig. 6b): The *Heddle* encoder processes exogenous static (e.g., adjacency matrix, elevation, etc.) and dynamic (e.g., time of day, day of week, weather, temperature, etc.) covariates to produce a representation of contextual information, which is then appended to the dataset.

The augmented dataset from the *Swatch* and *Heddle* are concatenated before passing into the *Loom* module that processes and examines the data in three different dimensional views before combining them using *P-KMV* followed by ELU, dropout and feedforward layers. The combined representations are finally passed through a linear forecasting layer to generate traffic predictions.

The three views of the data are:

- The *Weft*, $\mathcal{T} \in \mathbb{R}^{T \times E_T}$: The temporal embeddings of the data.
- The *WarP*, $\mathcal{P} \in \mathbb{R}^{N \times E_P}$: The spatial embeddings of the data.
- The *Draft*, $\mathcal{D} \in \mathbb{R}^{T \times N \times E}$: The spatio-temporal embeddings of the data.

$\mathcal{T}$, $\mathcal{P}$, and $\mathcal{D}$ are generated in the *WeftTer*, *WarPer*, and *Drafter* modules, respectively.

Encoder Modules

The *Swatch* (endogenous feature encoder)

The *Swatch* (Fig. 6a) is an encoder module that allows the *Weaver* to capture complex, non-linear spatio-temporal relationships in the data without relying on predefined structural assumptions.

The *Swatch* is simple by design. It consists a parameter tensor, $\hat{\mathcal{X}} \in \mathbb{R}^{N \times L}$ that learns L static latent features stratified by N nodes, followed by a MLP layer that projects $\hat{\mathcal{X}}$ to every t time

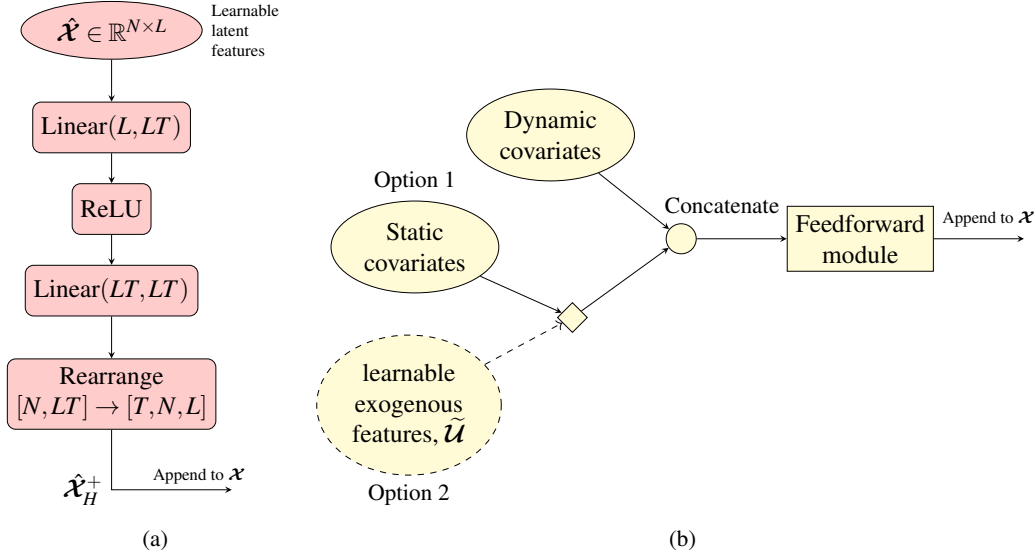
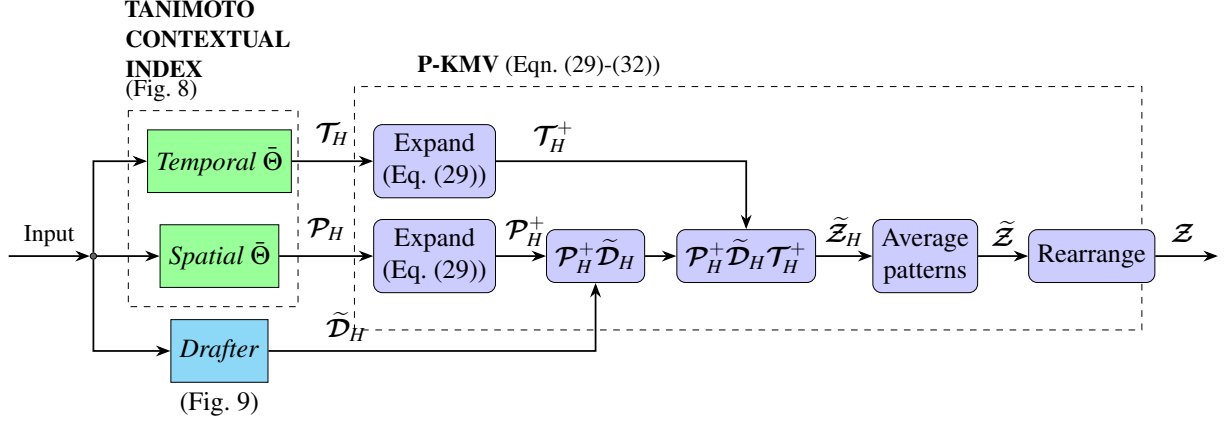


FIGURE 6: (a) Architecture of the *Swatch*, an encoder which uses a simple MLP to equip the *Weaver* with a capacity to learn static spatio-temporal associations between nodes in the spatio-temporal data. (b) Architecture of the *Heddle*, an exogenous feature encoder. Learnable exogenous embeddings can also be used concurrently or in place of static covariates, if necessary.

- 1 step, $\hat{\mathcal{X}}^+ \in \mathbb{R}^{T \times N \times L}$.
- 2 *The Heddle (exogenous feature encoder)*
- 3 The *Heddle* encoder (Fig. 6b) enhances the *Weaver*'s predictive power by efficiently combining static
- 4 and dynamic exogenous covariates into rich contextual representations. It uses a simple feedforward
- 5 network to merge static covariates (e.g., adjacency matrix) with dynamic ones (e.g., cyclical
- 6 temporal embeddings), offering better performance than direct concatenation or feature reduction
- 7 methods like reduced SVD. This approach allows fine-tuning of exogenous covariates' contribution
- 8 to model complexity via adjustable embedding size. If needed, a learnable exogenous feature
- 9 layer can replace static covariates, providing flexibility while maintaining minimal computational
- 10 overhead.
- 11 **The Loom Module**
- 12 The *Loom* module, depicted in Fig. 7, is the core component of the *Weaver*. It assimilates and
- 13 processes the input dataset using a Kronecker Matrix-Vector Identity product modified for parallel
- 14 Tensor operations.
- 15 **The WarPer and WefTer Modules**
- 16 The *WarP* ($\mathcal{P}$) and *WefT* ($\mathcal{T}$) are the spatial and temporal Tanimoto coefficients, respectively,
- 17 computed using the Tanimoto Coefficient module, with the only distinction being how the temporal
- 18 and spatial dimensions are rearranged prior to input.
- 19 The motivation behind using the Tanimoto Coefficient as the primary discriminator layer
- 20 comes the advantages discussed in (21):
- 21 • The Tanimoto coefficient is suitable for high dimensional and sparse data since the dot product

FIGURE 7: Architecture of the *Loom* module.

- 1 ensures only paired non-zero elements are utilized,
- 2 • It distinguishes vectors by angle and magnitude, which can produce richer latent feature
- 3 representations,
- 4 • It is backed by research from broad testing and application in the field of cheminformatics
- 5 and data mining.

6 The Tanimoto Coefficient Score Module (Tanimoto Module)

7 The Tanimoto Coefficient Score module (Fig. 8) computes the multi-head intradimensional Tanimoto
 8 coefficient. The output of this module is a tensor containing the association score on the interval
 9 $[-0.333, 1]$ between two twnsor slices along the instance-wise dimension (i.e., time step-wise or
 10 node-wise), where -0.333 indicating two slices with opposite directions but equal magnitudes,
 11 whereas 1 indicates two slices of equal magnitude and direction.

12 Let Θ denote the Tanimoto coefficient tensor, and we use the shorthand I_* in place of:
 13 $\{I_0 \times I_1 \times \dots \times I_{\eta-3}\}$. We can write $\mathcal{A} \in \mathbb{R}^{I_* \times B \times C}$ and $\mathcal{B} \in \mathbb{R}^{I_* \times D \times C}$, such that $\Theta \in \mathbb{R}^{I_* \times B \times D}$. Using
 14 element-wise division, the instance-wise Tanimoto coefficient tensor between $\mathcal{A}_{\dots:i}$ and $\mathcal{B}_{\dots:j}$ is
 15 computed as follows:

$$16 \text{ Tanimoto}(\mathcal{A}, \mathcal{B}) = \Theta(\mathcal{A}, \mathcal{B}) = \frac{\mathcal{A}\mathcal{B}^\top}{\bar{\mathcal{A}}^+ + (\bar{\mathcal{B}}^+)^\top - \mathcal{A}\mathcal{B}^\top}, \quad (11)$$

17 where each element $\Theta_{i_0 i_1 \dots i_{\eta-1}}$ of Θ is in the interval $[-\frac{1}{3}, 1]$. Here, $\bar{\mathcal{A}}^+$ and $\bar{\mathcal{B}}^+$ are feature-
 18 wise squared norms which are first computed and then expanded to the dimension of $(\mathcal{A}\mathcal{B}^\top)$ for
 19 element-wise division.

$$20 \begin{aligned} \bar{\mathcal{A}}^+ &= \text{Expand}(\xi = \eta - 1, \kappa = D) \left(\sum_k \mathcal{A}_{\dots,jk}^2 \right) = \text{Expand}(\xi = \eta - 1, \kappa = D) (\bar{\mathcal{A}}), \\ \bar{\mathcal{B}}^+ &= \text{Expand}(\xi = \eta - 1, \kappa = B) \left(\sum_k \mathcal{B}_{\dots,jk}^2 \right) = \text{Expand}(\xi = \eta - 1, \kappa = B) (\bar{\mathcal{B}}), \end{aligned} \quad (12)$$

21 where $\sum_k \mathcal{A}_{\dots,jk}^2$ computes the sum of squares along the last dimension i.e., $\eta - 1$, of $\mathcal{A}$, yielding
 22 $\bar{\mathcal{A}} \in \mathbb{R}^{I_* \times B \times 1}$ and likewise for $\mathcal{B}$ with $\bar{\mathcal{B}} \in \mathbb{R}^{I_* \times D \times 1}$.

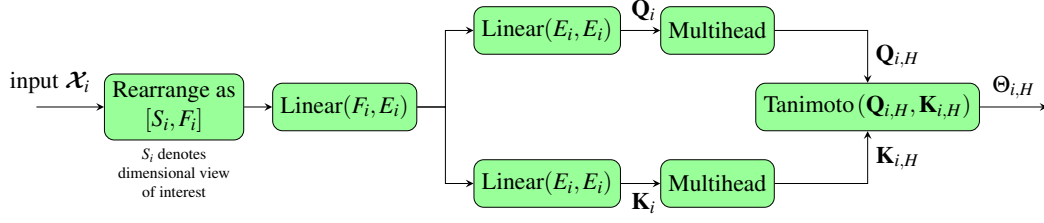


FIGURE 8: Architecture of the Tanimoto Coefficient Score module, where $i \in \{T, N\}$.

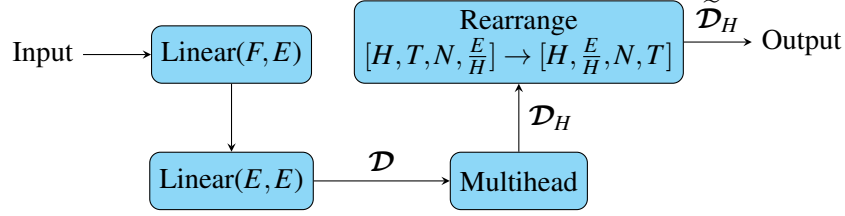


FIGURE 9: Architecture of the *Drafter*, the module that processes input spatio-temporal embeddings into the multi-head *Draft*, $\mathcal{D}_H$, and then the multi-head aligned-*Draft*, $\tilde{\mathcal{D}}_H$.

1 The formulation and output domain of the Tanimoto coefficient suggests that it is functionally
 2 both a normalizing layer and activation layer, which negates the need for a post-computation
 3 activation layer.

4 The *WarPer* and *WefTer* variants of the *Tanimoto module* differ only in dimensional rear-
 5 rangement to present the intended instance-wise stratification of spatio-temporal $\mathcal{X}$:

6 • **WarPer**: The input $\mathcal{X}_P$ is rearranged node-wise,

$$7 \quad \mathcal{X}_P = \left[\mathcal{X}_{[T \times N \times R]} \xrightarrow{T \leftrightarrow N} \mathcal{X}_{[N \times T \times R]} \xrightarrow{T \cdot R} \mathcal{X}_{[N \times TR]} \right], \quad (13)$$

8 • **WefTer**: The input $\mathcal{X}_T$ is rearranged time step-wise,

$$9 \quad \mathcal{X}_T = \left[\mathcal{X}_{[T \times N \times R]} \xrightarrow{N \cdot R} \mathcal{X}_{[T \times NR]} \right]. \quad (14)$$

10 The Drafter Module

11 The *Drafter* module (Fig. 9) processes the input dataset $\mathcal{X}$ into a multi-head latent representation of
 12 the data, the *Draft*, $\mathcal{D}_H \in \mathbb{R}^{H \times T \times N \times \frac{E}{H}}$, where E is the draft embedding size. $\mathcal{D}$ is then rearranged
 13 as the aligned-*Draft* $\tilde{\mathcal{D}}_H \in \mathbb{R}^{H \times \frac{E}{H} \times N \times T}$ in preparation for the P-KMV operation downstream.

$$14 \quad \mathcal{D} = \text{Linear}(E, E)(\text{Linear}(F, E)(\mathcal{X})), \quad (15)$$

$$15 \quad \mathcal{D}_H = \text{Multihead}(\mathcal{D}), \quad (16)$$

$$16 \quad \tilde{\mathcal{D}}_H = \left[\mathcal{D}_{H, [H, T, N, \frac{E}{H}]} \xrightarrow{T \leftrightarrow \frac{E}{H}} \mathcal{D}_{H, [H, \frac{E}{H}, N, T]} \right]. \quad (17)$$

17 The Shuttle

18 The *Shuttle* module applies the Kronecker Matrix-Vector product to combine spatial ($\mathcal{P}$) and tempo-
 19 ral (*WefTer*) association patterns, projecting onto latent features in $\mathcal{D}$ to generate comprehensive data
 20 representations. We'll first derive the generic expression for the Parallel Kronecker Matrix-Vector
 21 (P-KMV) product, which accelerates KMV computations. Then, we'll apply P-KMV to the $\mathcal{T}$, $\mathcal{P}$,
 22 and $\mathcal{D}$.

1 *The Parallel Kronecker Matrix-Vector Product (P-KMV)*

2 Starting from the classical definition of KMV from Eq. (2) restated below with $\mathbf{A} \in \mathbb{R}^{P \times Q}$, $\mathbf{B} \in \mathbb{R}^{R \times S}$,
 3 and $\mathbf{C} \in \mathbb{R}^{S \times Q}$:

$$4 \quad (\mathbf{A} \otimes \mathbf{B}) \text{vec}(\mathbf{C}) = \text{vec}(\mathbf{BCA}^\top). \quad (18)$$

5 substitute $\text{vec}(\mathbf{C}) = \mathbf{v}$ and $\mathbf{C} = \text{vec}^{-1}(\mathbf{v})$, such that $\mathbf{v} \in \mathbb{R}^{SQ \times 1}$ and $\text{vec}^{-1}(\mathbf{v}) \in \mathbb{R}^{S \times Q}$. Now
 6 rewrite Eq. (2) to get:

$$7 \quad (\mathbf{A} \otimes \mathbf{B})\mathbf{v} = \text{vec}(\mathbf{B} \text{vec}^{-1}(\mathbf{v})\mathbf{A}^\top). \quad (19)$$

8
 9 Now suppose that $\mathbf{v}_k$ is the k -th column vector of matrix $\mathbf{V} \in \mathbb{R}^{SQ \times K}$. We can indeed apply
 10 the column picture of matrix multiplication to $(\mathbf{A} \otimes \mathbf{B})\mathbf{V}$ to reinterpret Eq. (19) as an intermediate
 11 column-wise computation for $(\mathbf{A} \otimes \mathbf{B})\mathbf{V}$:

$$12 \quad (\mathbf{A} \otimes \mathbf{B})\mathbf{V} = [(\mathbf{A} \otimes \mathbf{B})\mathbf{v}_0 \mid (\mathbf{A} \otimes \mathbf{B})\mathbf{v}_1 \mid \cdots \mid (\mathbf{A} \otimes \mathbf{B})\mathbf{v}_{K-1}]. \quad (20)$$

13 By substituting Eq. (19) with proper indexing into Eq (20), we arrive at

$$14 \quad (\mathbf{A} \otimes \mathbf{B})\mathbf{V} = [\text{vec}(\mathbf{B} \text{vec}^{-1}(\mathbf{v}_0)\mathbf{A}^\top) \mid \text{vec}(\mathbf{B} \text{vec}^{-1}(\mathbf{v}_1)\mathbf{A}^\top) \mid \cdots \mid \text{vec}(\mathbf{B} \text{vec}^{-1}(\mathbf{v}_{K-1})\mathbf{A}^\top)], \quad (21)$$

15 which as established earlier, we have $\text{vec}^{-1}(\mathbf{v}_k) \in \mathbb{R}^{S \times Q}$.

16 For-loops could be used when calculating Eq. (21) for $k = 0, 1, 2, \dots, (K-1)$ when K is of
 17 moderate size. However, this is inefficient for high-dimensional tensors due to the exponential
 18 increase in number of column vectors across all dimensions. Hence, we propose the P-KMV
 19 formulation that is compatible with parallel tensor computations on infrastructures such as *NVidia*
 20 *CUDA* (22) to circumvent this and radically improve computational times.

21 To do this, we first substitute tensors of arbitrary dimensions into Eq. (20) and (21). Let
 22 $\mathcal{A} \in \mathbb{R}^{I_0 \times I_1 \times \cdots \times I_{\xi-3} \times P \times Q}$, $\mathcal{B} \in \mathbb{R}^{I_0 \times I_1 \times \cdots \times I_{\xi-3} \times R \times S}$, and $\mathcal{V} \in \mathbb{R}^{I_0 \times I_1 \times \cdots \times I_{\xi-3} \times SQ \times K}$. For legibility, we
 23 shall apply the shorthand: $I_* = \{I_0 \times I_1 \times \cdots \times I_{\xi-3}\}$:

$$24 \quad (\mathcal{A} \otimes \mathcal{B})\mathcal{V} = [(\mathcal{A} \otimes \mathcal{B})\mathbf{v}_0 \mid (\mathcal{A} \otimes \mathcal{B})\mathbf{v}_1 \mid \cdots \mid (\mathcal{A} \otimes \mathcal{B})\mathbf{v}_{K-1}] \\ = [\text{vec}(\mathcal{B} \text{vec}^{-1}(\mathbf{v}_0)\mathcal{A}^\top) \mid \text{vec}(\mathcal{B} \text{vec}^{-1}(\mathbf{v}_1)\mathcal{A}^\top) \mid \cdots \mid \text{vec}(\mathcal{B} \text{vec}^{-1}(\mathbf{v}_{K-1})\mathcal{A}^\top)], \quad (22)$$

25 where we now have $(\mathcal{A} \otimes \mathcal{B}) \in \mathbb{R}^{I_* \times RP \times SQ}$ and $\mathbf{v}_k \in \mathbb{R}^{I_* \times SQ \times 1}$. Applying the template from earlier,
 26 we can establish that $\text{vec}^{-1}(\mathbf{v}_k) \in \mathbb{R}^{I_* \times S \times Q}$.

27 The primary issue with parallelizing the formulation in Eq. (21) and (22) subsists in the
 28 $\text{vec}(\cdot)$ operation itself if we see $\text{vec}(\mathcal{B} \text{vec}^{-1}(\mathbf{v}_0)\mathcal{A}^\top)$ as an inseparable operation. Fortunately,
 29 Eq. (21) and (22) can be viewed as two consecutive but distinct tensor operations:

$$30 \quad \tilde{\mathcal{Z}}_k = \mathcal{B} \text{vec}^{-1}(\mathbf{v}_k)\mathcal{A}^\top : \tilde{\mathcal{Z}}_k \in \mathbb{R}^{I_* \times S \times Q}, \quad (23)$$

$$31 \quad \text{vec}(\tilde{\mathcal{Z}}_k) = \left[\tilde{\mathcal{Z}}_{k, [I_*, S, Q]} \xrightarrow{(S \times Q) \rightarrow (S \cdot Q \times 1)} \tilde{\mathcal{Z}}_{k, [I_*, SQ, 1]} \right], \quad (24)$$

32 it is apparent that the tensor rearrangement in Eq. (24) does not impart any additional information to
 33 $\tilde{\mathcal{Z}}_k$ that is not already extracted in Eq. (23). So for purposes in deep learning, Eq. (24) is cosmetic

1 and therefore an optional operation.

2 To parallelize Eq. (23), We require K replicates of $\mathcal{A}$ and $\mathcal{B}$ lined up with K number
 3 of $\text{vec}^{-1}(\mathbf{v}_k)$ for $k = 0, 1, 2, \dots, (K - 1)$. This can be achieved by applying $\text{Expand}(\cdot)$ before
 4 performing tensor multiplication:

$$\begin{aligned} \mathcal{A}^+ &= \text{Expand}(0, K)(\mathcal{A}) : \mathcal{A}^+ \in \mathbb{R}^{K \times I_* \times P \times Q}, \\ \mathcal{B}^+ &= \text{Expand}(0, K)(\mathcal{B}) : \mathcal{B}^+ \in \mathbb{R}^{K \times I_* \times R \times S}. \end{aligned} \quad (25)$$

6 K partitions of $\text{vec}^{-1}(\mathbf{v}_k)$ are then created from $\mathcal{V}$ as follows:

$$\tilde{\mathcal{V}} = \left[\mathcal{V}_{[I_*, SQ, K]} \xrightarrow{(I_*, SQ) \leftrightarrow K} \mathcal{V}_{[K, I_*, SQ]} \xrightarrow{SQ \rightarrow (S \times Q)} \mathcal{V}_{[K, I_*, S, Q]} \right]. \quad (26)$$

9 Finally, with Eq. (25) and (26), the Parallel KMV can be stated here:

$$\tilde{\mathcal{Z}} = \left(\mathcal{B}^+ \tilde{\mathcal{V}} (\mathcal{A}^+)^{\top} \right) : \tilde{\mathcal{Z}} \in \mathbb{R}^{K \times I_* \times R \times P}. \quad (27)$$

12 For completion, the form $\mathcal{Z} = (\mathcal{A} \otimes \mathcal{B})\mathcal{V} \in \mathbb{R}^{RP \times K}$ can be recovered by reversing the tensor
 13 rearrangement in Eq. (26). Although optional, but a spatio-temporal attention map can be obtained
 14 this way (see Eq. (35)):

$$\mathcal{Z} = \left[\tilde{\mathcal{Z}}_{K \times I_* \times R \times P} \rightarrow \tilde{\mathcal{Z}}_{K \times I_* \times RP} \rightarrow \tilde{\mathcal{Z}}_{I_* \times RP \times K} \right]. \quad (28)$$

17 Applying P-KMV to the spatio-temporal problem

18 Having derived the P-KMV in Eq. (25) through (27), the P-KMV can now be utilized to "weave"
 19 the spatio-temporal data views from the *WarPer* and *WeFTer* modules together, forming the core
 20 operation of the *Loom* that allows the *Weaver* to inspect and learn inter-dimensional interactions
 21 simultaneously in the dataset. Although some adjustments are made to accommodate the tensor
 22 dimensions in the problem, the overarching procedure remains intact.

23 This is done for the multi-head versions of $\mathcal{D}$, $\mathcal{T}$, and $\mathcal{P}$ without losing generality, since
 24 the mono-head case is simply $H = 1$, and we require multi-head activations to capture a variety of
 25 representations of spatial and temporal information from the data.

$$\begin{aligned} \mathcal{T}_H^+ &= \text{Expand} \left(1, \frac{E}{H} \right) (\mathcal{T}) : \mathcal{T}^+ \in \mathbb{R}^{H \times \frac{E}{H} \times T \times T}, \\ \mathcal{P}_H^+ &= \text{Expand} \left(1, \frac{E}{H} \right) (\mathcal{P}) : \mathcal{P}^+ \in \mathbb{R}^{H \times \frac{E}{H} \times N \times N}. \end{aligned} \quad (29)$$

27 Adjusting for the dimensionality of $\mathcal{D}_H$, the following rearrangements are made from $\mathcal{D}_H$,
 28 where E is the embedding size of the *Drafter* module. We shall refer to this rearranged $\mathcal{D}_H$ as the
 29 multi-head aligned-*Draft*, $\tilde{\mathcal{D}}_H$:

$$\tilde{\mathcal{D}}_H = \left[\mathcal{D}_{H, [H, T, N, \frac{E}{H}]} \xrightarrow{T \leftrightarrow \frac{E}{H}} \tilde{\mathcal{D}}_{H, [H, \frac{E}{H}, N, T]} \right]. \quad (30)$$

32 The multi-head spatio-temporal latent feature tensor are then computed as:

$$\tilde{\mathbf{Z}}_H = \left(\mathcal{P}_H^+ \tilde{\mathcal{D}}_H (\mathcal{T}_H^+)^T \right) : \tilde{\mathbf{Z}}_H \in \mathbb{R}^{H \times \frac{E}{H} \times N \times T}. \quad (31)$$

However, since the original spatio-temporal stratifications is desired, a final tensor rearrangement is performed and the multi-heads consolidated, then $\mathbf{Z}$ is passed through downstream layers before passing it to the final Linear($E, 1$) layer to generate forecasted traffic measure values:

$$\mathbf{Z} = \left[\tilde{\mathbf{Z}}_{H, [H, \frac{E}{H}, N, T]} \xrightarrow{H \cdot E} \tilde{\mathbf{Z}}_{[E, N, T]} \xrightarrow{E \leftrightarrow T} \mathbf{Z}_{[T, N, E]} \right]. \quad (32)$$

EXPERIMENTS

Core Research Questions

The objective of our experiments are to:

- Evaluate the *Weaver*'s performance in several encoder configurations, and benchmark the performance against two spatio-temporal Transformer models: STTN and STAEformer.
- Investigate the *Weaver* model's performance on a dataset with low missing rates (PEMS-BAY, 0.02%) and moderately high missing rates (METR-LA, 8.11%).
- Conduct a preliminary inspection of the *Weaver*'s temporal and spatial activation maps during three distinct traffic periods during a weekday: peak hours (12:00-14:00), regular hours (09:00-11:00), and off-peak night hours (02:00-04:00). A reduced *Weaver* model with 4 activation heads is used on the PEMS-BAY dataset for simpler visual inspection.

Experimental Setup

All models were trained and tested on NVidia Tesla T4 GPUs via Google Colab, using the Torch spatio-temporal Library (TSL) (23) for data preparation and processing. Datasets were chronologically split into 70:20:10 for training, validation, and testing. Benchmark model hyperparameters were chosen based on the values used in their respective papers, or using values for the closest network size when specific dataset parameters were unavailable.

Model Complexities

Table 1 summarizes the precursory analysis of model complexities and the required exogenous features for models to perform learning and forecasting.

Inspection of the time complexities suggests that the STTN has lower time complexity than the *Weaver* when $L \ll N$ and $L \ll T$, whereas the STAEformer is on-par with *Weaver* when $L = 1$. For space complexity, the STTN is on-par with *Weaver* when $L = 1$, whereas the *Weaver* has lower space complexity than the STAEformer when $L \ll N$ and $L \ll T$.

Datasets

The datasets are sourced from the TSL library (23), including the preparations and sequence slicing procedures. Two datasets were selected for the benchmark tests: PEMS-BAY, and METR-LA. For each baseline model, no additional exogenous covariates are included in the training data unless stated in their respective papers.

Data preprocessing

The input spatio-temporal data $\mathbf{X} \in \mathbb{R}^{T \times N \times 1}$ consists of T timesteps, N nodes, and one type of traffic measure.

The features of the raw data, $\mathbf{X}^{\text{raw}}$, are first standardized by subtracting the mean values

Model	Time complexity	Space complexity	Learnable embedding	Positional embedding	Graph connectivity	Temporal embedding
Weaver	$\mathcal{O}(NT(N+T))$	$\mathcal{O}(N^2+T^2)$	Yes	No	Optional	Optional
Vanilla Transformer	$\mathcal{O}(LN^2T^2)$	$\mathcal{O}(LN^2T^2)$	N/A	Yes	N/A	N/A
STTN	$\mathcal{O}(L(N^2+T^2))$	$\mathcal{O}(L(N^2+T^2))$	Yes	Yes	Yes	Optional
STAEformer	$\mathcal{O}(LNT(N+T))$	$\mathcal{O}(LNT(N+T))$	Yes	Yes	No	Yes

TABLE 1: Time and space complexity comparison between models, where L is the number of attention layers. The exogenous features required for models are also shown.

and dividing by the respective standard deviations:

$$\mathcal{X}_{ijk} = \frac{\mathcal{X}_{ijk}^{\text{raw}} - \mu_k}{\sigma_k}, \quad (33)$$

where $\mathcal{X}_{ijk}$ represents the standardized value at the i -th timestep, j -th node, and k -th feature, $\mathcal{X}_{ijk}^{\text{raw}}$ represents the original, unstandardized value at the i -th timestep, j -th node, and k -th feature, μ_k is the mean of the k -th feature, and σ_k is the standard deviation of the k -th feature. The standardization is reversed before the training and validation losses are calculated against the target data.

Exogenous covariates

Traffic datasets typically include two exogenous covariates: time data (date and time of each sample) and node connectivity information (adjacency matrix).

For the *Weaver*, time of day and day of week were cyclically encoded to a continuous range [0,1] to capture their periodicity.

$$\Psi(\mathbf{u}) = \begin{bmatrix} \sin(2\pi\omega\mathbf{u}) \\ \cos(2\pi\omega\mathbf{u}) \end{bmatrix}, \quad (34)$$

where $\mathbf{u}$ is the periodical exogenous covariate and $\frac{1}{\omega}$ the period of oscillation.

STAEformer requires time of day normalized on [0,1) and integer-encoded days of week from 0 to 6. The STTN does not use temporal encodings. Adjacency matrices from the TSL library (23) are used by some *Weaver* configurations and STTN models. However, the STAEformer use learned spatial features instead. This information is also reflected in Table 1.

Model Setup and Hyperparameters

The *Weaver*'s hyperparameters are summarized in Table 2.

We used the STTN implementation from (5) with hyperparameters based on (24). For PEMS-BAY, we used three spatial-temporal blocks with single attention and 64 feature channels. For METR-LA, we used one block with the same configuration, based on the similar-sized PeMSD6 dataset. The forward expansion multiplier was set to 2 for METR-LA and 1 for PEMS-BAY.

For STAEformer, the implementation from (25) was used, whereas the hyperparameter values prescribed in (6) were used. The feature embedding dimension is 24 and the spatio-temporal adaptive embedding dimension is 80, with 3 layers and 4 heads for both spatial and temporal transformers.

TABLE 2: Summary of *Weaver* hyperparameter values and configurations used in experiments

Model config.	Num. heads, H	Time embed dim., E_T	Space embed dim., E_P	Draft embed dim., E	FFW hidden dim., E_{FFW}	Heddle embed dim., E_h	Num. learned endo. embs.	Num. learned exog. embs.	Temporal exog.	adj. matrix
<i>Weaver</i>	16	48	48	48	128	-	-	-	-	-
+ <i>Swatch</i> (S-16LE)						0	16	0	✗	✗
+ <i>Heddle</i> (H-T/A)						32	0	0	✓	✓
+ <i>Heddle</i> (H-16EX/T)						32	0	16	✓	✗

1 Training and Optimizer Configurations

2 All models were trained on the Mean Absolute Error (MAE) loss function with 50 epochs and a
3 batch size of 32. For the optimizer, *Weaver*, and STAEformer uses the *Adam* optimizer and the
4 StepLR scheduler, with initial step size 0.001, a 5 epoch update step, and a step size reduction factor
5 of 0.75. The STTN model uses the RMSprop optimizer with the StepLR scheduler, initial step size
6 0.001, 5 epoch update, and a reduction factor of 0.70 to be more consistent with the setup described
7 in (24).

8 RESULTS

9 Model Benchmarks

10 We evaluated the performance of several state-of-the-art spatio-temporal traffic forecasting models
11 on two benchmark datasets: PEMS-BAY 2017 Speed and METR-LA 2012 Speed. The models
12 under comparison included STTN, STAEformer, and variants of our proposed *Weaver* architecture.
13 Performance was assessed using the average RMSE, MAPE, and MAE for 15-, 30-, and 60-minute
14 prediction horizons, with 95% confidence intervals reported to account for variability across multiple
15 trials. The results of these benchmarks are compiled in Table 3. For the *Weaver*, they collectively
16 suggest that the model performs well in both datasets with low and moderate percentage of missing
17 data.

TABLE 3: Summary of results showing average performance metrics (RMSE, MAPE, MAE) for 15-, 30-, and 60-min predictions with 95% error margins in subscript parenthesis. Includes model parameters and average seconds per training epoch. Best results per dataset are bold with asterisks. Metrics are averaged over at least 5 trials with different random seeds, potentially differing from original papers.

Data (Missing rate)	Model (Num. trials)	Avg. RMSE _(95%Err.)			Avg. MAPE _(95%Err.)			Avg. MAE _(95%Err.)			Num. Params.
		15 min	30 min	60 min	15 min	30 min	60 min	15 min	30 min	60 min	
PEMS-BAY 2017 Speed (0.02%)	STTN (21)	3.10 _(±0.01)	4.05 _(±0.01)	4.95 _(±0.01)	3.34 _(±0.01)	4.44 _(±0.01)	5.67 _(±0.01)	1.54 _(±0.01)	1.92 _(±0.01)	2.34 _(±0.01)	541k
	STAEformer (5)	3.00 _(±0.08)	3.99 _(±0.16)	4.66 _(±0.13)	3.14 _(±0.16)	4.09 _(±0.17)	4.90 _(±0.23)	1.46 _(±0.09)	1.80 _(±0.09)	2.08 _(±0.09)	1.4M
	<i>Weaver</i> (-)	-	-	-	-	-	-	-	-	-	-
	+ <i>S-16LE</i> (15)	2.96 _(±0.01)	3.92 _(±0.02)	4.61 _(±0.03)	2.97* _(±0.02)	3.91 _(±0.03)	4.73 _(±0.03)	1.38* _(±0.00)	1.71 _(±0.00)	2.01 _(±0.01)	364k
	+ <i>H-16EXT</i> (29)	2.95* _(±0.01)	3.81* _(±0.01)	4.41* _(±0.01)	2.97* _(±0.01)	3.82 _(±0.01)	4.56* _(±0.02)	1.38* _(±0.00)	1.67* _(±0.00)	1.93* _(±0.00)	639k
	+ <i>H-T/A</i> (31)	2.96 _(±0.01)	3.82 _(±0.02)	4.44 _(±0.05)	2.98 _(±0.01)	3.81* _(±0.03)	4.56* _(±0.07)	1.38* _(±0.00)	1.67* _(±0.01)	1.93* _(±0.02)	683k
METR-LA 2012 Speed (8.11%)	STTN (13)	5.29* _(±0.06)	6.42* _(±0.10)	7.80 _(±0.18)	7.76* _(±0.28)	9.84 _(±0.42)	12.76 _(±0.51)	2.81* _(±0.08)	3.37 _(±0.11)	4.12 _(±0.13)	140k
	STAEformer (5)	5.68 _(±0.04)	6.61 _(±0.07)	7.55 _(±0.15)	8.05 _(±0.21)	9.41 _(±0.16)	10.80 _(±0.42)	2.99 _(±0.08)	3.31 _(±0.06)	3.66 _(±0.06)	1.3M
	<i>Weaver</i> (-)	-	-	-	-	-	-	-	-	-	-
	+ <i>S-16LE</i> (21)	5.75 _(±0.02)	6.77 _(±0.03)	7.70 _(±0.04)	7.95 _(±0.07)	9.55 _(±0.07)	11.19 _(±0.10)	2.92 _(±0.01)	3.32 _(±0.01)	3.74 _(±0.01)	245k
	+ <i>H-16EXT</i> (10)	5.70 _(±0.03)	6.57 _(±0.03)	7.39 _(±0.04)	7.84 _(±0.05)	9.12* _(±0.05)	10.56 _(±0.08)	2.88 _(±0.01)	3.20 _(±0.01)	3.54 _(±0.01)	385k
	+ <i>H-T/A</i> (10)	5.66 _(±0.02)	6.55 _(±0.02)	7.36* _(±0.04)	7.80 _(±0.06)	9.12* _(±0.08)	10.52* _(±0.14)	2.86 _(±0.01)	3.18* _(±0.01)	3.50* _(±0.02)	412k

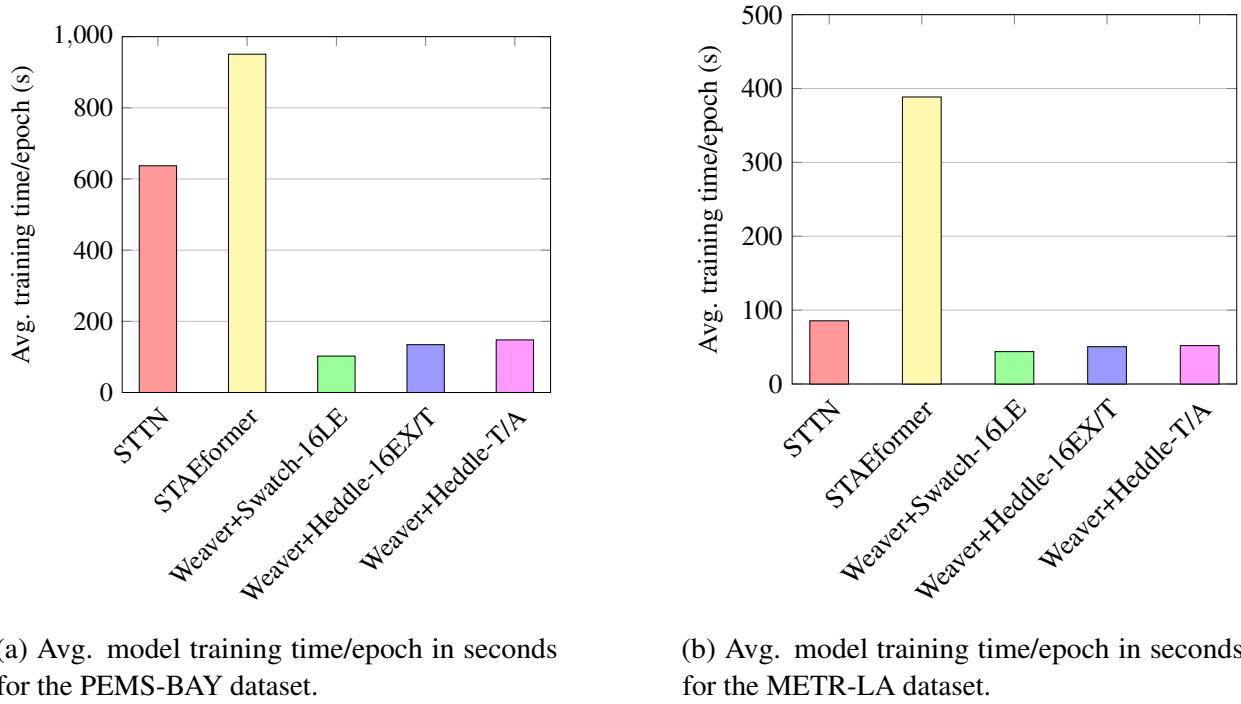


FIGURE 10: Average model training time per epoch in seconds.

The results demonstrate that the *Weaver* variants, particularly H-16EX/T and H-T/A, exhibit competitive or superior performance across most metrics on both datasets. On the PEMS-BAY dataset, *Weaver* variants consistently outperformed other architectures, especially for longer-term predictions. For instance, the H-16EX/T variant achieved the lowest RMSE, MAPE, and MAE of 4.41, 4.56, and 1.93 respectively, for 60-minute predictions, compared to 4.95, 5.67, and 2.34 for STTN and 4.57, 4.80, and 2.09 for STAEformer.

For METR-LA, while STTN exhibited strong performance for short-term forecasts with (RMSE, MAPE, MSE) of (5.29, 7.76, 2.81) for 15-minute predictions and RMSE of 6.42 for 30-minute, *Weaver* variants excelled in longer-term predictions. For example, The H-T/A configuration achieved the lowest RMSE of 7.36 for 60-minute predictions, outperforming both STTN (7.80) and STAEformer (7.55).

According to the average training time per epoch (s) shown in Fig. 10, the *Weaver* variants demonstrate superior training efficiency. On PEMS-BAY, S-16LE, H-16EX/T, and H-T/A variants required 102.31, 134.42, and 147.80 seconds per epoch, respectively, which is significantly less than STTN (637.16s) and STAEformer (950.70s). Similarly, for METR-LA, *Weaver* variants needed 43.83-51.95 seconds per epoch, compared to 85.54s for STTN and 388.56s for STAEformer. A limitation of this comparison is that the STTN and STAEformer hyperparameters are taken from the original papers without modifications. Future comparisons should include a rigorous hyperparameter search for time-optimal configurations to ensure better comparisons.

Activation Maps

A smaller *Weaver* model is trained on the PEMS-BAY dataset to produce the activation maps for easier visual inspection. The following hyperparameters were used: 20 epochs, 4 activation heads,

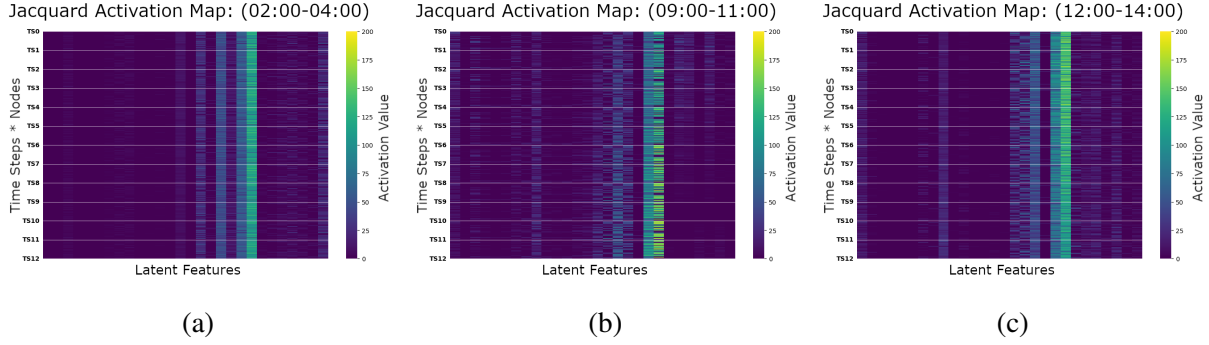


FIGURE 11: The Jacquard activation map (JAM) for $(\mathcal{T} \otimes \mathcal{P})\mathcal{D}$, showing 3 distinct but gradual phase shifts from night off-peak hours (a), to morning regular hours (b), then to mid-day peak hours (c). The nodes in each time step (TS) are separated by white lines. The color scale is on the range [0,200]. (a) JAM for night off-peak hours (02:00-04:00), (b) JAM for mid-day regular hours (09:00-11:00), (c) JAM for noon rush hours (12:00-14:00)

1 $E_T = E_P = E = 28$, and using the *Heddle* encoder with exogenous temporal data and 16 learned
 2 exogenous features. All other hyperparameters follow the experiments, and no adjacency matrix
 3 was used.

4 The MAE, MAPE, and MSE values evaluated on the test set at 15, 30, and 60 minutes are,
 5 respectively: 1.41, 1.72, 1.99 (MAE); 3.09%, 3.99%, 4.77% (MAPE); and 3.02, 3.92, 4.53 (RMSE).

6 *The Jacquard Activation Map (JAM)*

7 The Jacquard Activation Map, $\mathcal{J} \in \mathbb{R}^{TN \times E}$ (named after the "Jacquard machine", a loom attachment
 8 that automates weaving complex fabric patterns, not to be confused with the Jaccard index) contains
 9 the interactions of latent features in $\mathcal{D}$ with $\mathcal{T}$ and $\mathcal{P}$. It is obtained by rearranging Eq. (32) and
 10 passing it through an ELU layer, which means the JAM is not defined on a closed interval:

$$11 \quad \mathcal{J} = \text{ELU} \left[\mathcal{Z}_{[T,N,E]} \xrightarrow{T \cdot N} \mathcal{Z}_{[TN,E]} \right] = \text{ELU}[(\mathcal{T} \otimes \mathcal{P})\mathcal{D}]. \quad (35)$$

12 The JAM in Fig. 11 shows the activation maps from PEMS-BAY on May 31, 2017 (Wednes-
 13 day) in three traffic states during morning hours (02:00-04:00), regular hours (09:00-11:00), and
 14 peak hours (12:00-14:00). The x-axis corresponds to each latent feature in the Draft, $\mathcal{D}$ from the
 15 *Drafter*, with each column representing one latent feature. The y-axis shows all the nodes grouped
 16 by time step (TS0, TS1,...,TS11). The color scale of the JAM ranges from 0 to 200, with yellow
 17 indicating higher values and dark blue lower values.

18 These maps provide insight into which features are most influential in the model's predictions
 19 at different times. Early morning hours (02:00-04:00) (Fig. 11a) show high activation concentrated
 20 in a few specific latent features, suggesting the model relies on a limited set of key indicators during
 21 typically quiet traffic periods. Regular traffic hours (09:00-11:00) (Fig. 11b) display a more diverse
 22 activation pattern across a wider range of feature columns, indicating that the model incorporates a
 23 broader set of factors as traffic patterns become more complex. Peak hours (12:00-14:00) (Fig. 11c)
 24 exhibit a concentrated version of Fig. 11b, with more nodes reaching high activation ranges in the
 25 brightest column and less overall variety in activated columns. This likely reflects the model's
 26 focus on the most critical features during peak traffic conditions when certain patterns dominate the
 27 network.

Overall, some latent features maintain high activations across all time periods, as evidenced by the consistently bright columns in all three subfigures. This suggests their fundamental importance in traffic prediction regardless of the time of day. Conversely, other features show time-dependent activation, highlighting how the model learned contextual patterns based on changing traffic conditions.

These Jaccard Activation Maps reveal our model’s adaptability in encoding different latent features as traffic patterns evolve. The progression from sparse activation in early morning to complex, concentrated patterns during peak hours demonstrates the model’s capacity to capture varying traffic complexities. This dynamic feature encoding allows for context-aware predictions, potentially improving traffic forecasting accuracy throughout the day.

Tanimoto Tartan Activation Maps (TAM)

Tanimoto Tartan Activation Maps (TAM), named after plaid-like tartan patterns, display intra-temporal associations in temporal ($T \times T$) and spatial ($N \times N$) dimensions, computed using the instance-wise Tanimoto Coefficient from Eq. (11). TAMs visualize intra-dimensional association patterns learned by the *Weaver* for predictions.

Temporal TAMs (T-TAM) are shown in local scale (Fig. 12) to highlight activation patterns, and in global scale (Fig. 13) on $[-0.333, 1]$ range for true-scale activations. Spatial TAMs (P-TAM) are similarly presented in local scale (Fig. 14) to highlight activation patterns, and in global scale (Fig. 15). The PiYG color map uses pink for low values, white for neutral, and green for high values.

The top row of subfigures represent TAMs for early morning traffic (02:00-04:00), middle for regular traffic (09:00-11:00) and bottom for peak hour traffic (12:00-14:00). Conversely, each column represents outputs from a single activation head. The rows and columns of the TAMs corresponds to time steps for T-TAMs, and nodes for P-TAMs.

The T-TAMs in Fig. 13 consistently display three distinct zones (pink, white, and green) across all activation heads, which may suggest a soft clustering of T-nodes. This tri-color pattern persists throughout the day with variations in zone size, polarity, and intensity. Early morning hours show scattered patterns, indicating lower time step associations. Regular hours exhibit more apparent zones with gentler gradients, while peak hours display sharper, concentrated blocks, suggesting high interconnections between adjacent time steps.

Notable observations include two polarity-flips in head 3 between Fig. 12a, 12b, and 12c, and a clockwise rotation of patterns in head 4. While the implications of these shifts are not yet clear, they reveal the *Weaver*’s diverse approaches to encoding temporal associations.

The P-TAMs in Fig. 14 reveal intricate $N \times N$ node-to-node relationships across different times of day. These maps show predominantly pink and white patterns with sporadic green highlights. The patterns vary subtly across time periods and attention heads, suggesting the model adapts its spatial understanding as traffic conditions change. Notably, the maps maintain a consistent structure of vertical and horizontal lines, hinting at persistent spatial relationships in the road network. Further investigation of these patterns could yield insights into the model’s spatial reasoning.

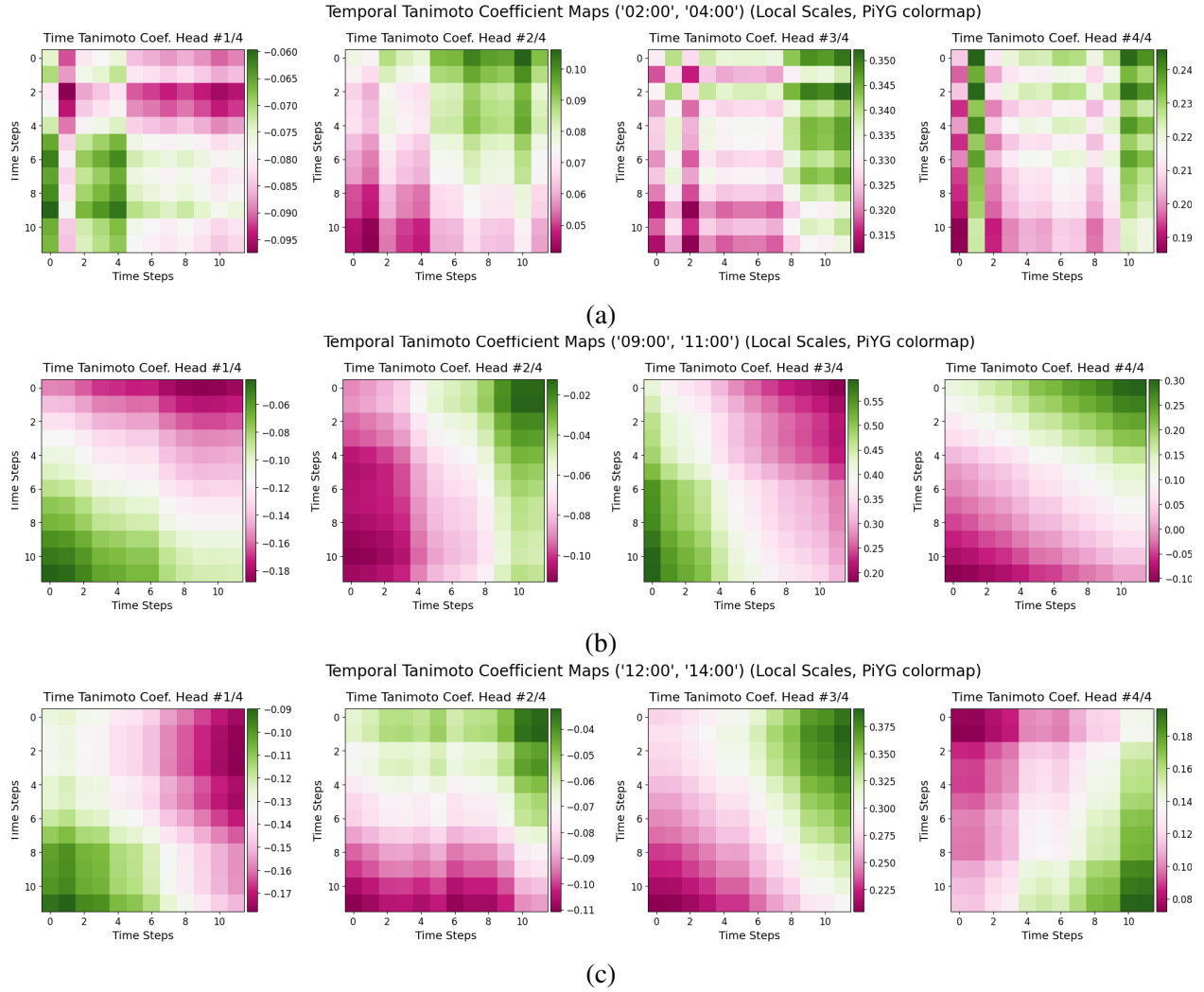


FIGURE 12: The $T \times T$ four-head temporal Tanimoto Tartan activation map (T-TAM), *corresponding by column*, and uses a **local color scale**. (a) temporal TAM for night off-peak hours (02:00-04:00), (b) temporal TAM for mid-day regular hours (09:00-11:00), (c) temporal TAM for noon rush hours (12:00-14:00)

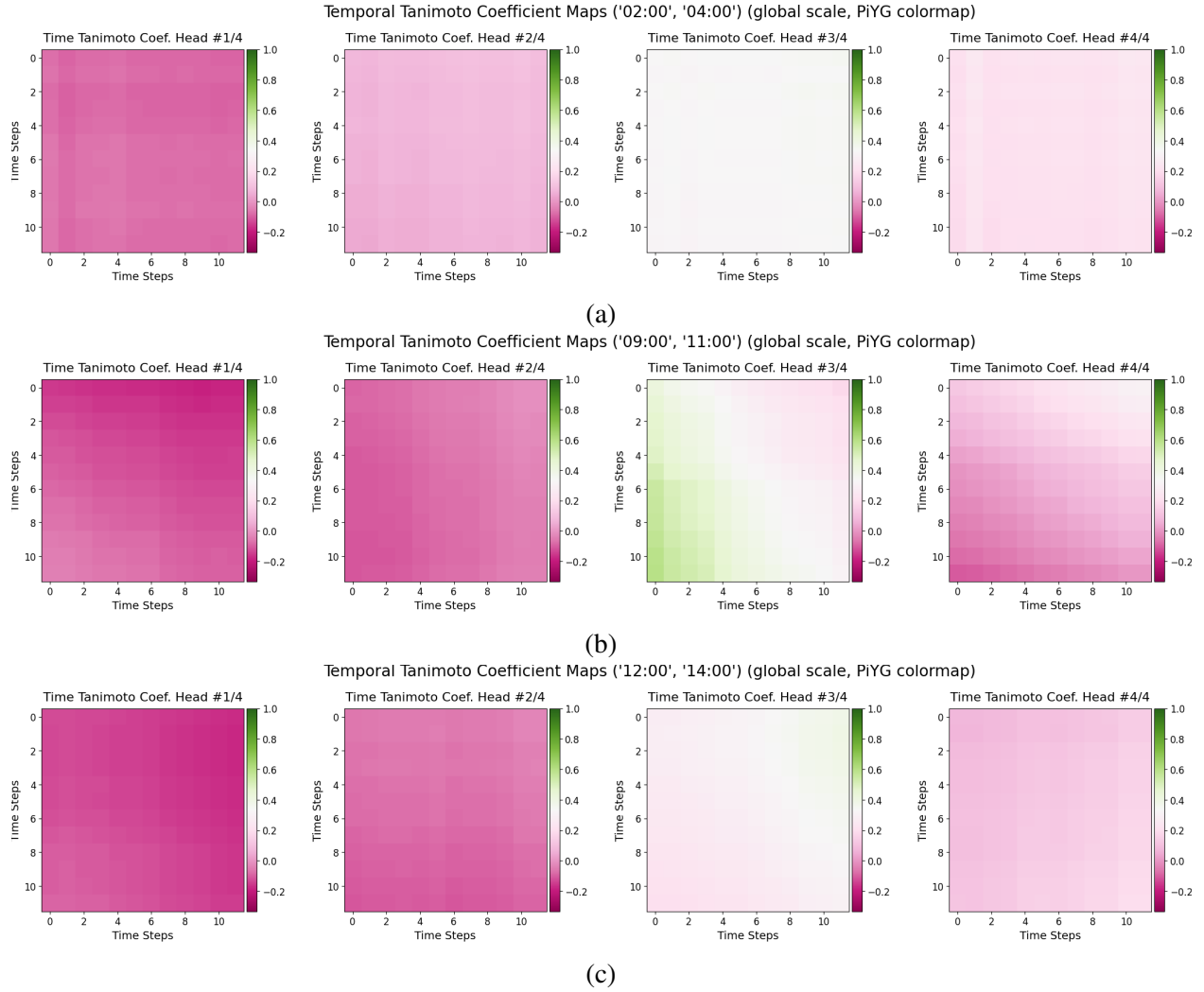


FIGURE 13: The $T \times T$ four-head temporal Tanimoto Tartan activation map (T-TAM), *corresponding by column*, and uses a **global color scale** on the interval $[-0.333, 1]$. (a) temporal TAM for night off-peak hours (02:00-04:00), (b) temporal TAM for mid-day regular hours (09:00-11:00), (c) temporal TAM for noon rush hours (12:00-14:00)

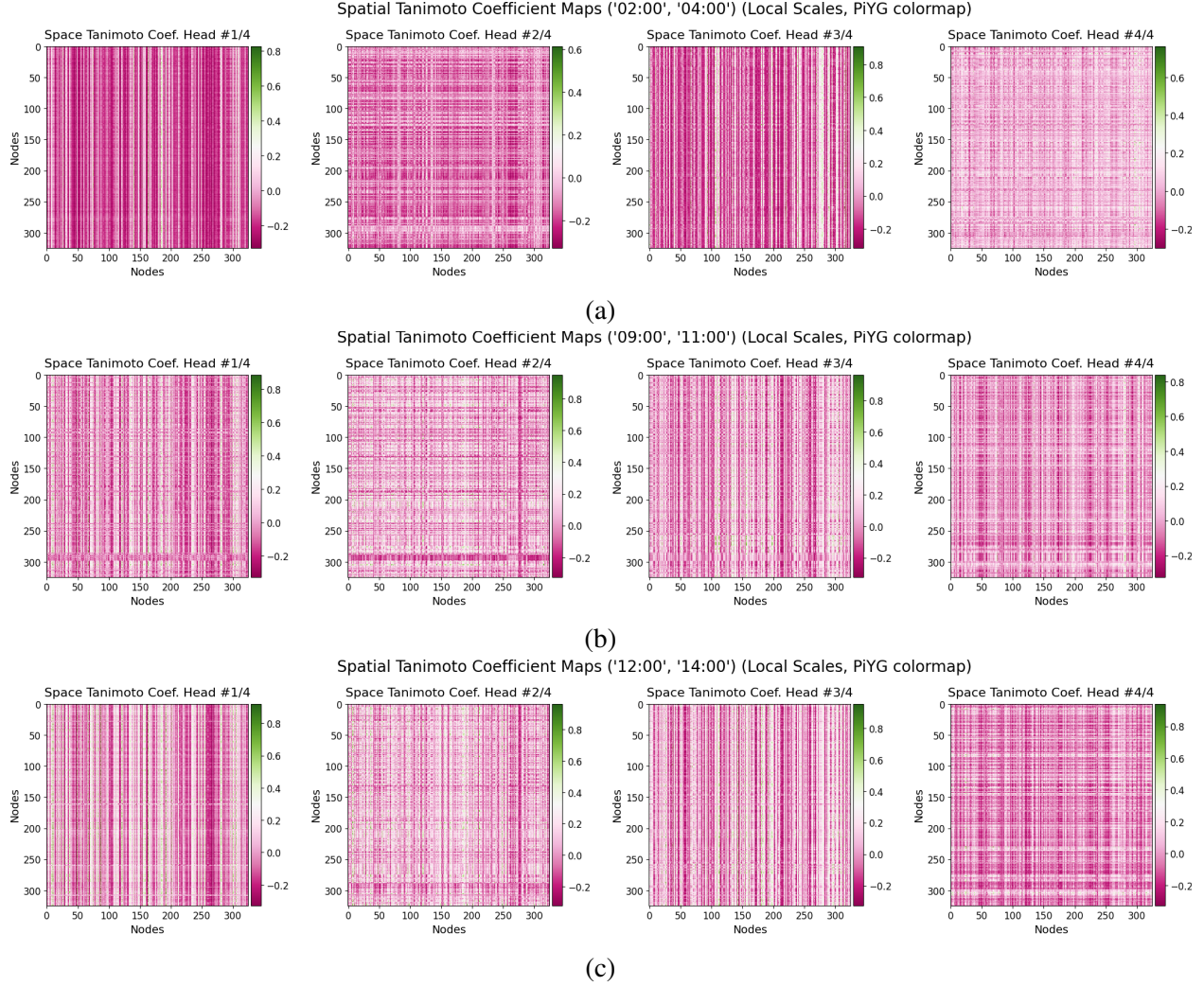


FIGURE 14: The $N \times N$ four-head spatial Tanimoto Tartan activation map (P-TAM), *corresponding by column*, and uses a **local color scale**. (a) spatial TAM for night off-peak hours (02:00-04:00), (b) spatial TAM for mid-day regular hours (09:00-11:00), (c) spatial TAM for noon rush hours (12:00-14:00)

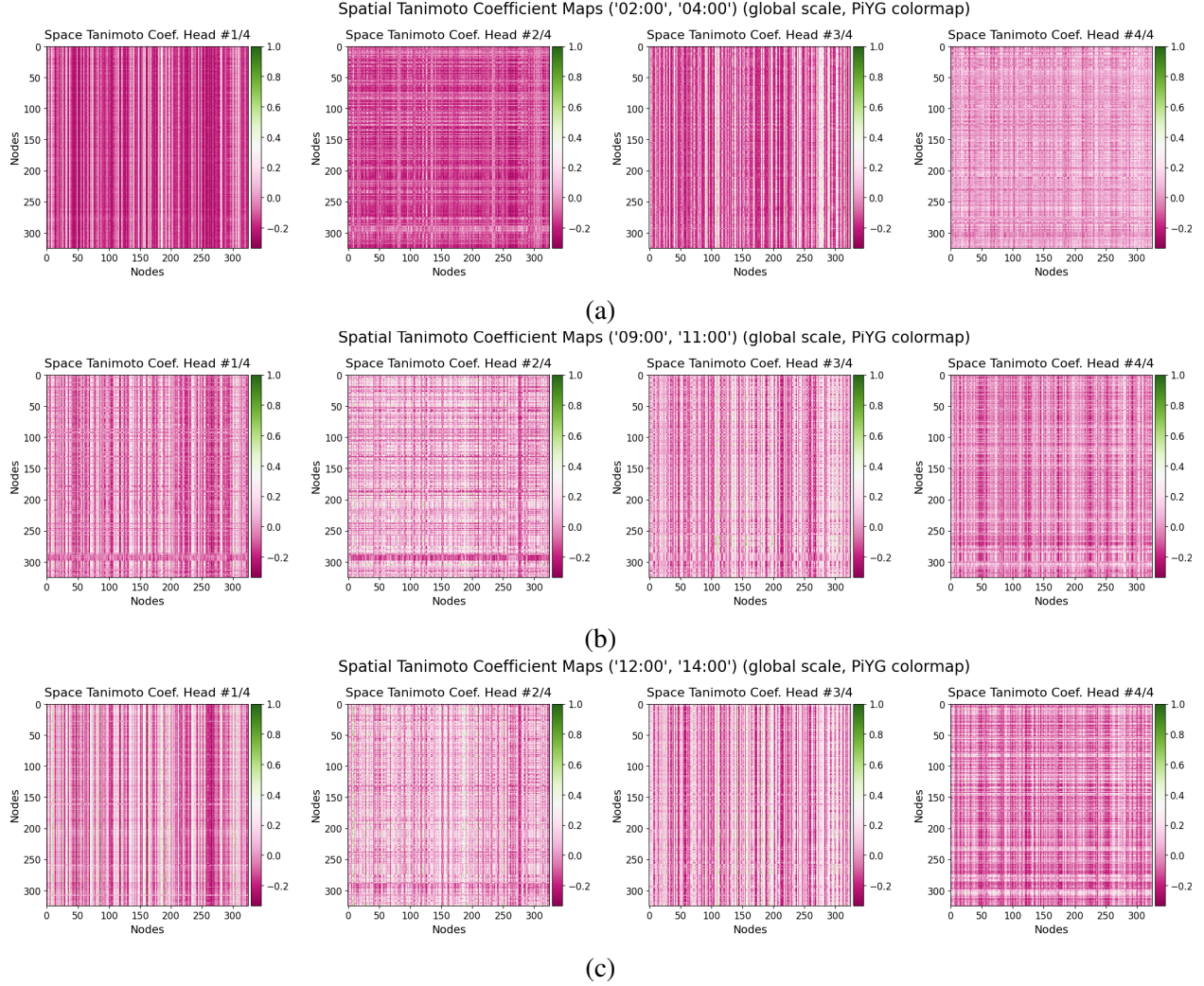


FIGURE 15: The $N \times N$ four-head spatial Tanimoto Tartan activation map (P-TAM), *corresponding by column*, and uses a **global color scale** on the interval $[-0.333, 1]$. **(a)** spatial TAM for night off-peak hours (02:00-04:00), **(b)** spatial TAM for mid-day regular hours (09:00-11:00), **(c)** spatial TAM for noon rush hours (12:00-14:00)

1 CONCLUSION

2 In this work, we proposed the *Weaver*, a novel deep learning architecture for learning and predicting
3 spatio-temporal data, and established the *Weaver* as a promising new entrant in the field of spatio-
4 temporal modeling. We conducted model training and forecasting tests using the PEMS-BAY and
5 METR-LA datasets and determined that the *Weaver* offers several key advantages. The *Weaver*'s
6 simple, non-recurring, and modular design offers easy construction, customization, and intuitive
7 usage. Unlike Transformers, it doesn't rely on positional embeddings, making its performance
8 independent of sequence order representation mechanisms—a key advantage for spatio-temporal
9 forecasting. The Swatch encoder enables learning static representations of spatio-temporal patterns
10 without temporal or connectivity information, while remaining flexible enough to utilize such data
11 when available.

12 The *Weaver* leverages parallel computing to achieve quick training and deployment, a small
13 memory footprint, and robust forecasting performance. The *Weaver*'s structure, lacking large-scale
14 repeating modules found in Transformers or recurrent networks, facilitates straightforward mapping
15 of activation maps and hidden states to inputs, potentially enhancing interpretability. Finally, the
16 intra-dimensional association maps produced with the Tanimoto coefficient may offer interesting
17 traffic pattern insights, warranting further investigation into potential interpretability,

18 The results look promising, and we intend to pursue several concepts and tasks in our
19 future work. First, we plan to benchmark the *Weaver* against other spatio-temporal forecasting
20 models beyond the STTN and STAEformer, while also testing it against various spatio-temporal
21 datasets within and beyond traffic forecasting. We will investigate the *Weaver*'s interactions with
22 exogenous covariates to design specialized protocols for incorporating specific types of information,
23 such as graph connectivity and temporal data, to improve model performance and interpretability.
24 Additionally, we aim to explore the mechanisms behind how the *Weaver* utilizes the Tanimoto
25 coefficient to encode intra-dimensional associations and how this information is used in P-KMV
26 calculations. To further our understanding, we will perform extensive ablative studies to map the
27 effects of varying hyperparameter values and encoders on the *Weaver*'s performance. We also intend
28 to develop techniques to enforce activation structure and consistency within the Jacquard activation
29 map across all training tasks, enabling straightforward comparisons between training outcomes
30 and improving interpretability. Lastly, we will work on developing better tools and techniques
31 to visualize and interpret model activation maps, enhancing our ability to analyze and refine the
32 *Weaver*'s performance.

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1. **Claude 3.5, ChatGPT-4o/4:** LaTeX formatting, TIKZ diagram creation, and preliminary proofreading (feedback on grammar, clarity, academic voice, and citations).
2. **GitHub Co-pilot:** Python code assistance (error diagnosis, syntax, library suggestions, and structuring).

The authors recognize the limitations of AI tools including errors, biases, and knowledge gaps. All content has been thoroughly reviewed and verified by the authors, who take full responsibility for the manuscript's accuracy and originality.

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